EWB AWS INTERNSHIP PROJECT REPORT

AWS VPC Peering • Multi-VPC Network Architecture on AWS

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Project Title	AWS VPC Peering	**Region**	ap-southeast-2 (Asia Pacific – Sydney)
Completion Date	November 2, 2026	**Status**	Completed

1 Architecture Diagram

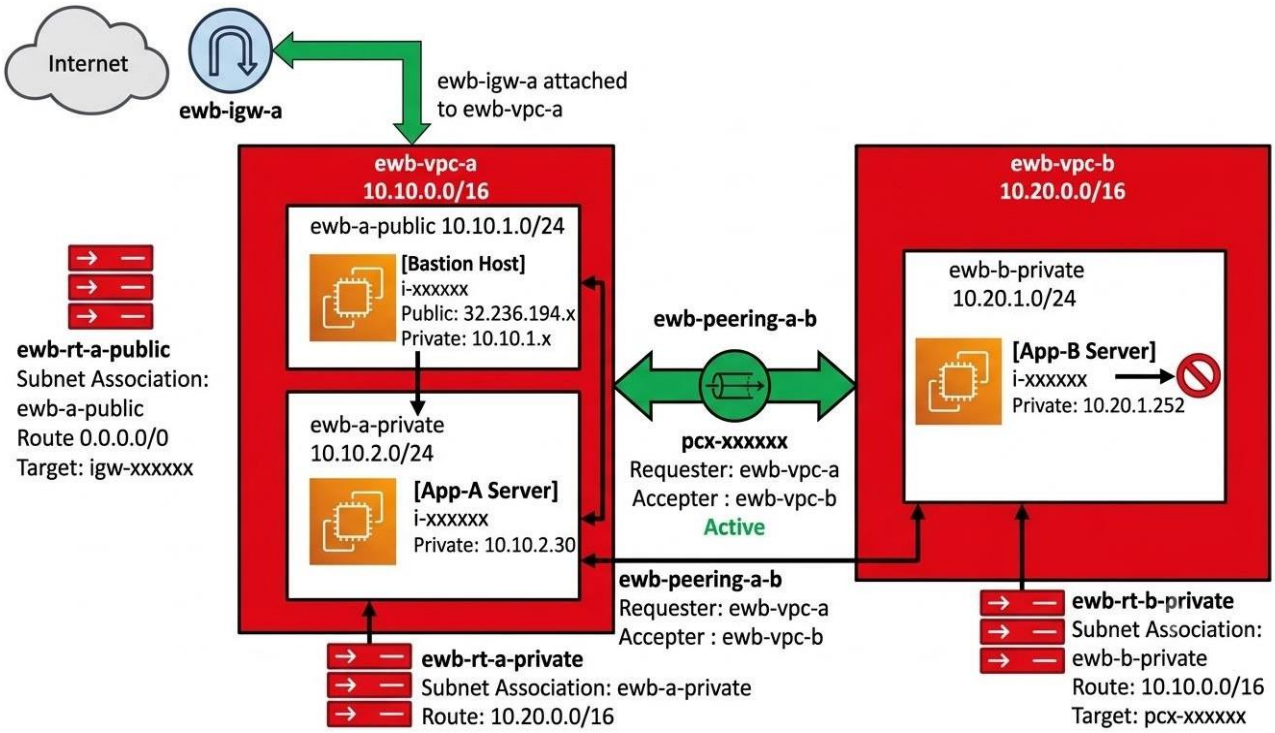


Figure 1. Two-VPC architecture: ewb-vpc-a (bastion + App-A, public/private subnets) peered with ewb-vpc-b (App-B, private subnet) via pcx-xxxxxx.

2 Resource Inventory & Network Parameters

VPC CIDR Blocks	`ewb-vpc-a: 10.10.0.0/16 | ewb-vpc-b: 10.20.0.0/16`
Subnet CIDR Ranges	`ewb-a-public: 10.10.1.0/24 | ewb-a-private: 10.10.2.0/24 | ewb-b-private: 10.20.1.0/24`
Bastion Host (ewb-ec2-bastion)	`Public IP: 32.236.194.219 | Private IP: 10.10.1.x | Subnet: ewb-a-public`
App-A Server (ewb-ec2-app-a)	`Private IP: 10.10.2.30 | Subnet: ewb-a-private | Security Group: ewb-sg-app-a`

App-B Server (ewb-ec2-app-b)	`Private IP: 10.20.1.252 | Subnet: ewb-b-private | Security Group: ewb-sg-app-b`
VPC Peering Connection	`pcx-xxxxxx — Requester: ewb-vpc-a, Accepter: ewb-vpc-b, Status: Active`
Internet Gateway	`ewb-igw-a (attached to ewb-vpc-a)`

3 Route Tables & Subnet Associations

Route Table	Associated Subnet	Destination	Target
`ewb-rt-a-public`	ewb-a-public (10.10.1.0/24)	`10.10.0.0/16` `0.0.0.0/0`	`Local` `ewb-igw-a`
`ewb-rt-a-private`	ewb-a-private (10.10.2.0/24)	`10.10.0.0/16` `10.20.0.0/16`	`Local` `pcx-xxxxxx`
`ewb-rt-b-private`	ewb-b-private (10.20.1.0/24)	`10.20.0.0/16` `10.10.0.0/16`	`Local` `pcx-xxxxxx`

4 Security Groups & Inbound Rules

Security Group	Protocol / Port	Source	Purpose
`ewb-sg-bastion`	`SSH (22)`	0.0.0.0/0	Admin terminal access to the bastion/jump host
`ewb-sg-app-a`	`SSH (22)`	10.10.1.0/24 (Bastion subnet)	SSH access from the bastion host only
`ewb-sg-app-a`	`All ICMP – IPv4`	10.20.0.0/16 (VPC-B range)	Allow ping replies from App-B over the peering link
`ewb-sg-app-b`	`SSH (22)`	10.10.0.0/16 (VPC-A range)	SSH access from hosts in VPC-A
`ewb-sg-app-b`	`All ICMP – IPv4`	10.10.0.0/16 (VPC-A range)	Allow ping requests from App-A over the peering link

5 Forward & Return Path Explanation

Forward Path

App-A (10.10.2.30) sends an ICMP Echo Request to App-B (10.20.1.252). Route table **ewb-rt-a-private** directs the 10.20.0.0/16 destination to the peering connection **pcx-xxxxxx**. Traffic crosses the AWS backbone into VPC-B, is routed by **ewb-rt-b-private**, and is permitted by App-B's security group (**ewb-sg-app-b**) ICMP inbound rule.

Return Path

App-B replies with an ICMP Echo Reply to 10.10.2.30. VPC-B's route table **ewb-rt-b-private** directs the 10.10.0.0/16 destination back over **pcx-xxxxxx** into VPC-A. The reply is evaluated statefully against App-A's security group (**ewb-sg-app-a**) and reaches App-A, achieving 0% packet loss.

6 Test Evidence & Validation Records

Test 1 — Baseline Failure Proof (Initial ICMP Block)	**FAIL (expected)**
Host	`ec2-user@ip-10-10-2-30`
Command	`ping 10.20.1.252 (prior to adding the ICMP inbound rule)`
Result	**100% packet loss.**
Explanation	Confirmed that VPC Peering routes alone are insufficient: although the route existed, App-B's security group still blocked the ICMP Echo Request until the inbound rule was added.

Test 2 — Forward & Reverse Ping Success Proof	**PASS**
Host	`ec2-user@ip-10-10-2-30`
Command	`ping 10.20.1.252 (after adding the All-ICMP-IPv4 rule)`
Result	**7 packets transmitted, 7 received, 0% packet loss.**
Explanation	Adding the ICMP inbound rule enabled full round-trip communication across the peering tunnel, confirming correct forward- and return-path operation.

Test 3 — App-B SSH Proof & Key Management	**PASS**
Host	`ec2-user@ip-10-10-1-x (Bastion)` → `ec2-user@ip-10-10-2-30 (App-A)`
Command	`chmod 400 bastion-key.pem && ssh -i bastion-key.pem ec2-user@10.10.2.30`
Result	**Successful SSH login.**
Explanation	Permissions were set to 400 (read-only, owner) on bastion-key.pem to satisfy SSH client requirements, enabling secure jump-box access to the private instance.

Test 4 — Controlled ICMP Failure & Recovery Validation	**PASS**
Command	`Temporarily removed the ICMP inbound rule from ewb-sg-app-b, then re-ran ping 10.20.1.252; reconfigured the rule and re-tested`
Result	**Rule removed → 100% packet loss (instant). Rule restored → 0% packet loss (instant).**
Explanation	Validated that connectivity is governed in real time by the security group state, confirming the firewall — not the route — was the controlling factor.

7 Cleanup Evidence

1. Terminated all EC2 instances.
2. Deleted the VPC Peering Connection.
3. Detached and deleted the Internet Gateway ewb-igw-a from ewb-vpc-a.
4. Deleted custom VPCs ewb-vpc-a and ewb-vpc-b, along with their associated subnets and route tables, to prevent unauthorized billing.

Summary

Through this hands-on project, I gained practical knowledge of multi-VPC architecture design, non-overlapping CIDR block allocation, bastion jump-host configuration, SSH key permission security, and bidirectional route table propagation over VPC Peering. I also gained hands-on experience troubleshooting network connectivity and security group firewall rules.